

A Multispecies Size-Spectrum Approach to Ecosystem-Based Fisheries Management in Iberian Atlantic Waters

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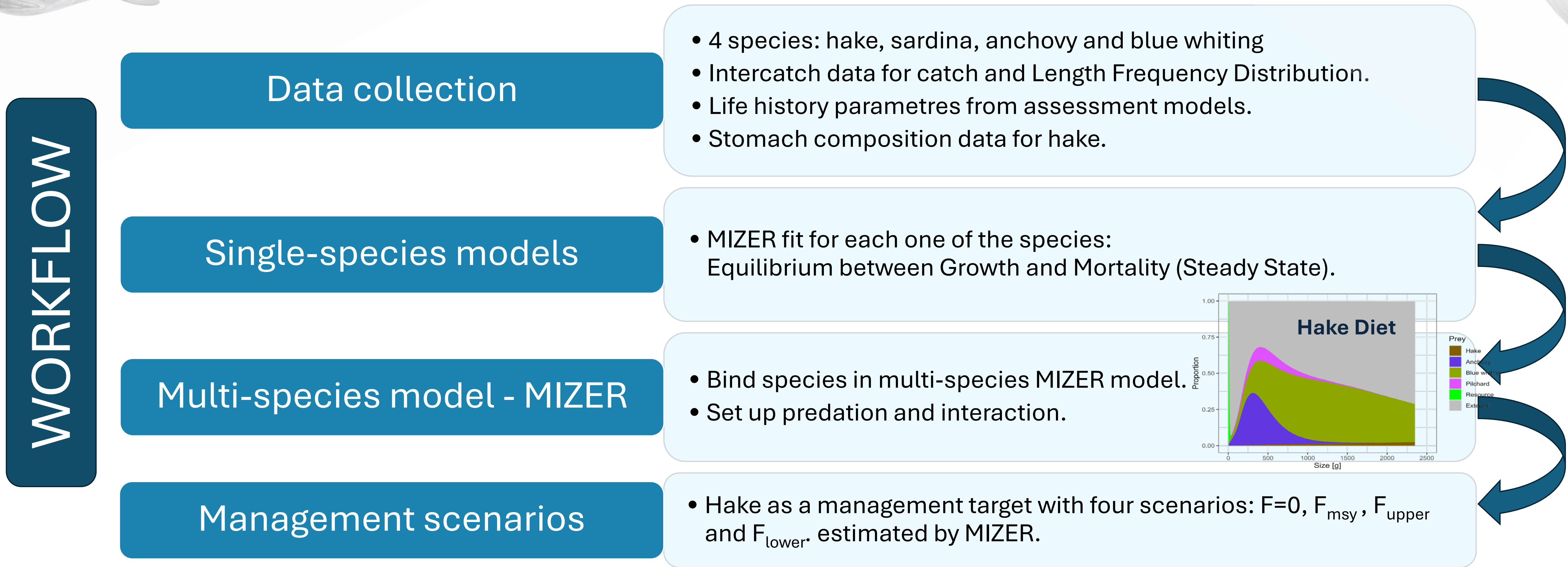
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 <https://mervex-group.github.io/MERVEX/proof/FRESCO.html>

Marine ecosystems are increasingly affected by multiple stressors such as climate change and fishing pressure, challenging traditional single-species management approaches. **Ecosystem-Based Fisheries Management (EBFM)** seeks to incorporate species interactions, environmental variability and cumulative human impacts into decision-making processes. The **MIZER** multi-species size-spectrum modelling framework offers a balance between ecological realism and practical applicability by representing trophic interactions through the community size structure. We are developing a multispecies ecosystem model for the Atlantic fish demersal community, using hake as a key predator and management target.

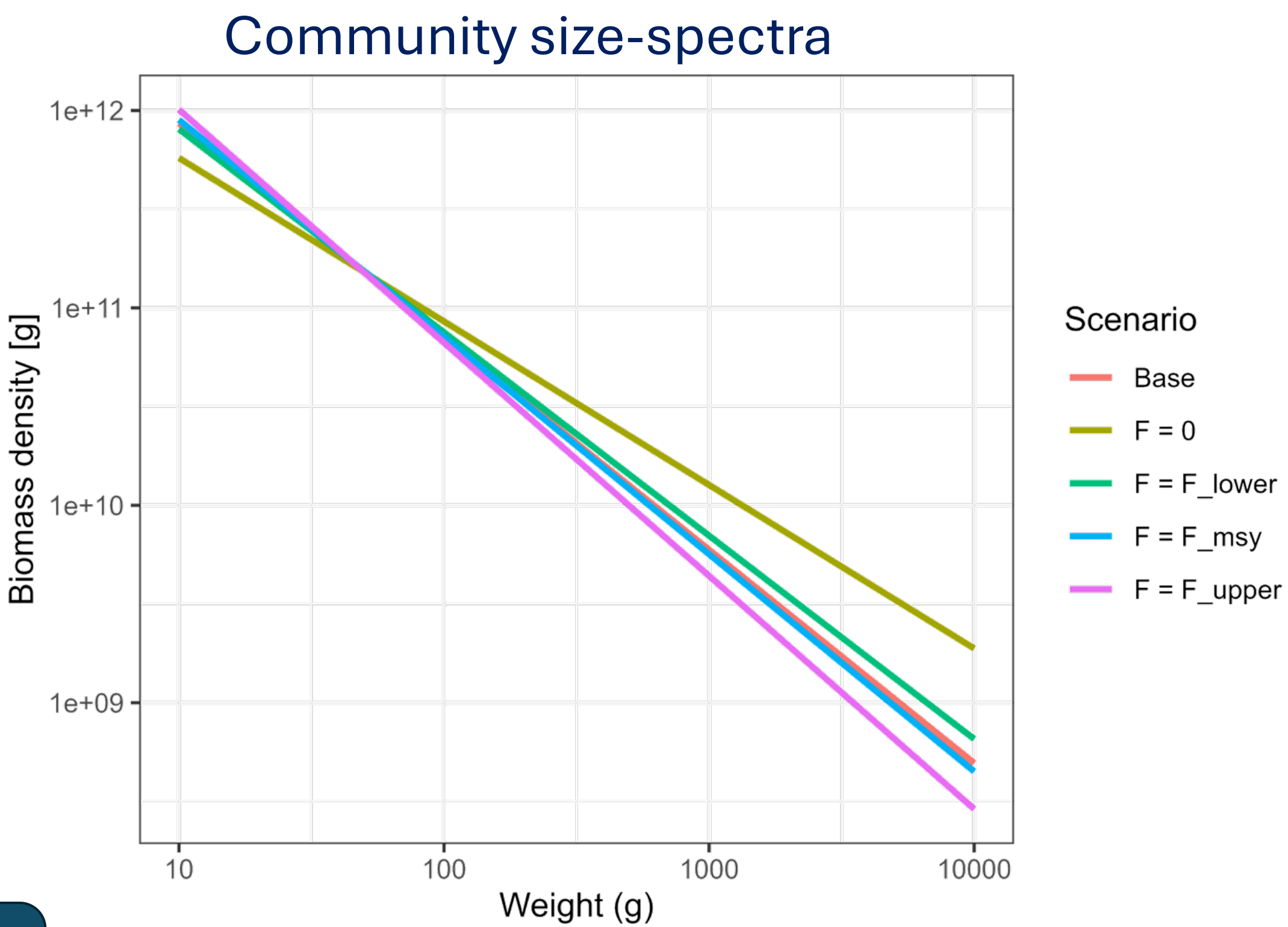
OBJECTIVE: To evaluate the impact of hake TAC management on its prey species' abundance.



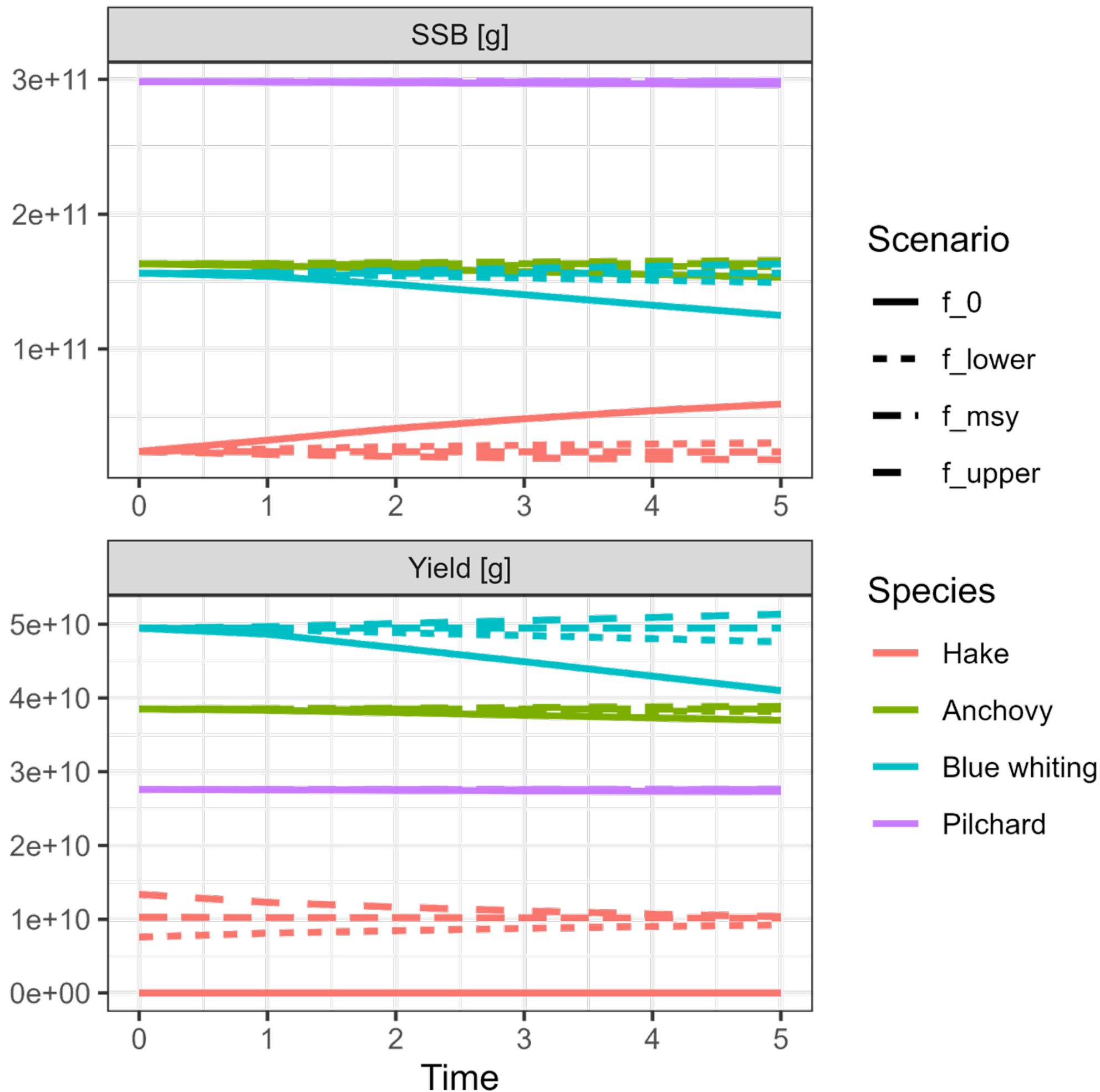
RESULTS

Short term forecast - catch options table with impact on prey species

Basis	Hake Catch (2025)	Hake MIZER F (2025)	Hake SSB (2026)	% hake SSB change*	Blue whiting SSB (2026)	% blue whiting SSB change*	Anchovy SSB (2026)	% anchovy SSB change*	Sardine SSB (2026)	% sardine SSB change*
$F = F_{msy}$	10232	0,081	23826	-0,32	156163	-0,01	163187	+0,01	298365	+0,002
$F = F_{lower}$	8105	0,060	27464	+6,26	154267	-0,88	162347	-0,30	298171	-0,04
$F = F_{upper}$	12286	0,107	20252	-7,40	158128	+0,87	164058	+0,33	298563	+0,04
$F = 0$	0	0	41120	+27,17	147834	-3,96	159499	-1,41	297491	-0,19



Five year forecast – constant F



CONCLUSIONS

- MIZER is a size spectra model with additional capabilities allowing the dynamic tracking of different species in the community.
- As expected, community size spectra slope decreases along fishing intensity.
- The impact of hake TAC on prey abundance is limited in the short time because of the high comparative abundance between hake and preys. The magnitude of prey responses differs among species, reflecting their distinct trophic links with hake.
- However, the impact is higher for blue whiting, the main prey in the area, and is also higher in the mid term (5 year). Anchovy and sardine show weak responses.
- Short-term ecosystem effects remain limited compared with the direct effect of fishing mortality on hake.
- The results support incorporating trophic interactions into hake management strategies. Future work will focus in adding more species with fleet interactions (mixed fisheries) as well of adding ecosystem indicators

This communication is part of the research project FRESCO: PID2022-1402900B-I00 funded by MCIN/AEI/10.13039/501100011033/ and by "ERDF A way of making Europe"; co-funded by the European Union through the European Maritime Fisheries and Aquaculture Fund (EMFAF) within the National Program of collection, management and use of data in the fisheries sector and support for scientific advice regarding the Common Fisheries Policy; and MeCCAM: funded by the European Union under Grant Agreement N° 101181571